

Table 1. Instrument-strength calibration

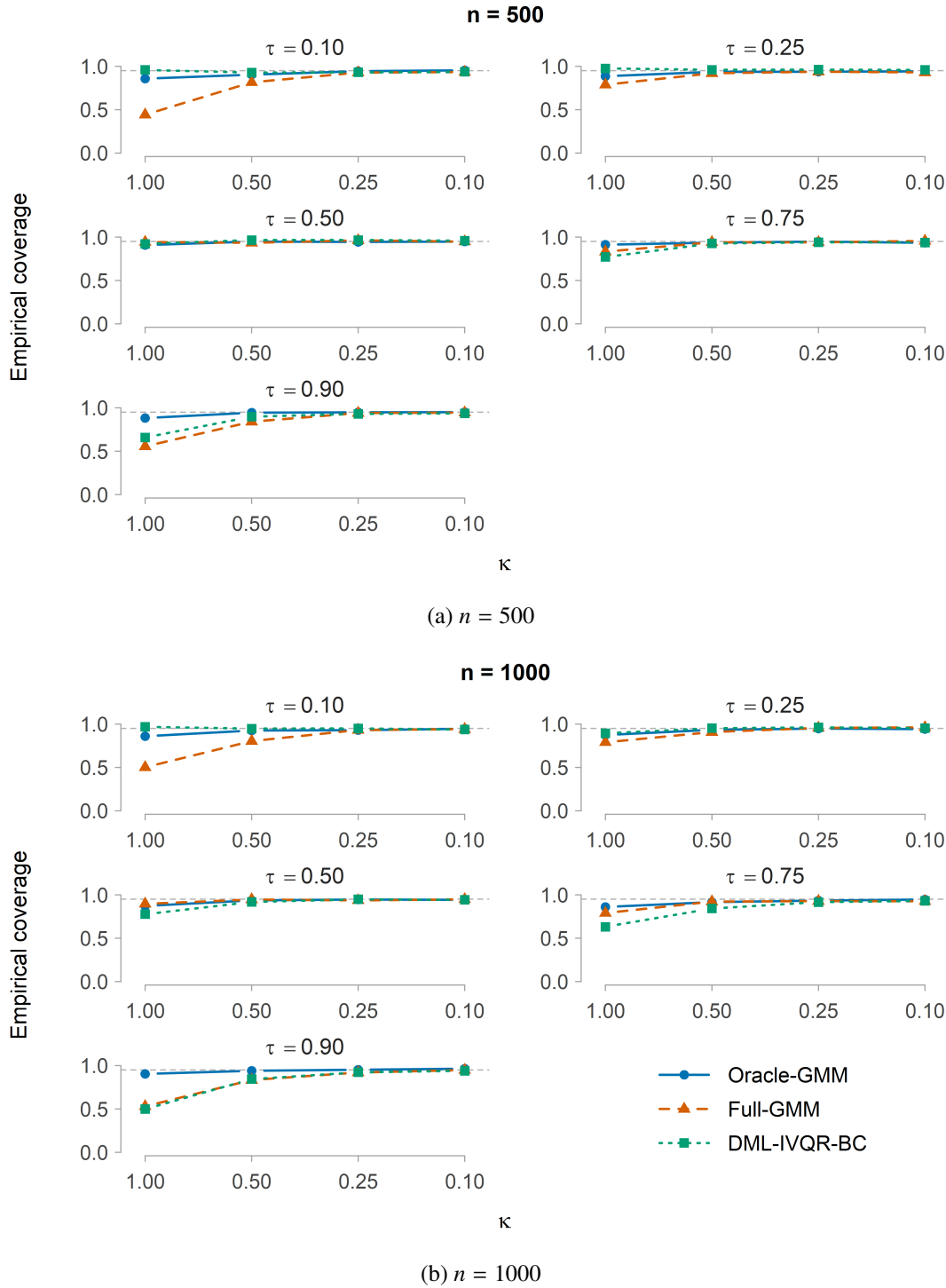
n	κ	Median F	F : p10–p90	Median partial R^2	Partial R^2 : p10–p90
500	1.00	101.17	85.93–126.68	0.294	0.261–0.342
500	0.50	18.71	11.99–27.95	0.071	0.047–0.103
500	0.25	4.74	1.74–8.65	0.019	0.007–0.034
500	0.10	1.22	0.24–3.65	0.005	0.001–0.015
1000	1.00	212.82	185.34–239.50	0.301	0.273–0.327
1000	0.50	38.65	30.13–53.60	0.073	0.058–0.098
1000	0.25	9.85	5.34–16.15	0.020	0.011–0.032
1000	0.10	2.04	0.38–5.42	0.004	0.001–0.011

Note: Descriptive Monte Carlo strength diagnostics from the preserved 100-replication calibration. The reported dispersion ranges are the 10th and 90th percentiles. These statistics are not formal weak-instrument classification thresholds.

Table 2. Point-estimation performance at the median quantile, $n = 1000$

κ	Estimator	Bias	MAE	RMSE
1.00	Oracle-GMM	0.007	0.047	0.072
1.00	Full-GMM	0.007	0.072	0.100
1.00	DML-IVQR-BC	0.070	0.079	0.104
0.50	Oracle-GMM	0.015	0.093	0.134
0.50	Full-GMM	0.014	0.151	0.199
0.50	DML-IVQR-BC	0.059	0.119	0.160
0.25	Oracle-GMM	0.045	0.281	0.431
0.25	Full-GMM	0.029	0.372	0.500
0.25	DML-IVQR-BC	-0.003	0.306	0.448
0.10	Oracle-GMM	-0.044	0.735	0.945
0.10	Full-GMM	-0.012	0.840	1.018
0.10	DML-IVQR-BC	-0.195	0.779	0.991

Note: Each cell uses 500 Monte Carlo replications. The frozen sign convention is $\text{Bias} = \alpha_0 - \hat{\alpha}$.

**Figure 1.** Empirical coverage across instrument-strength designs.

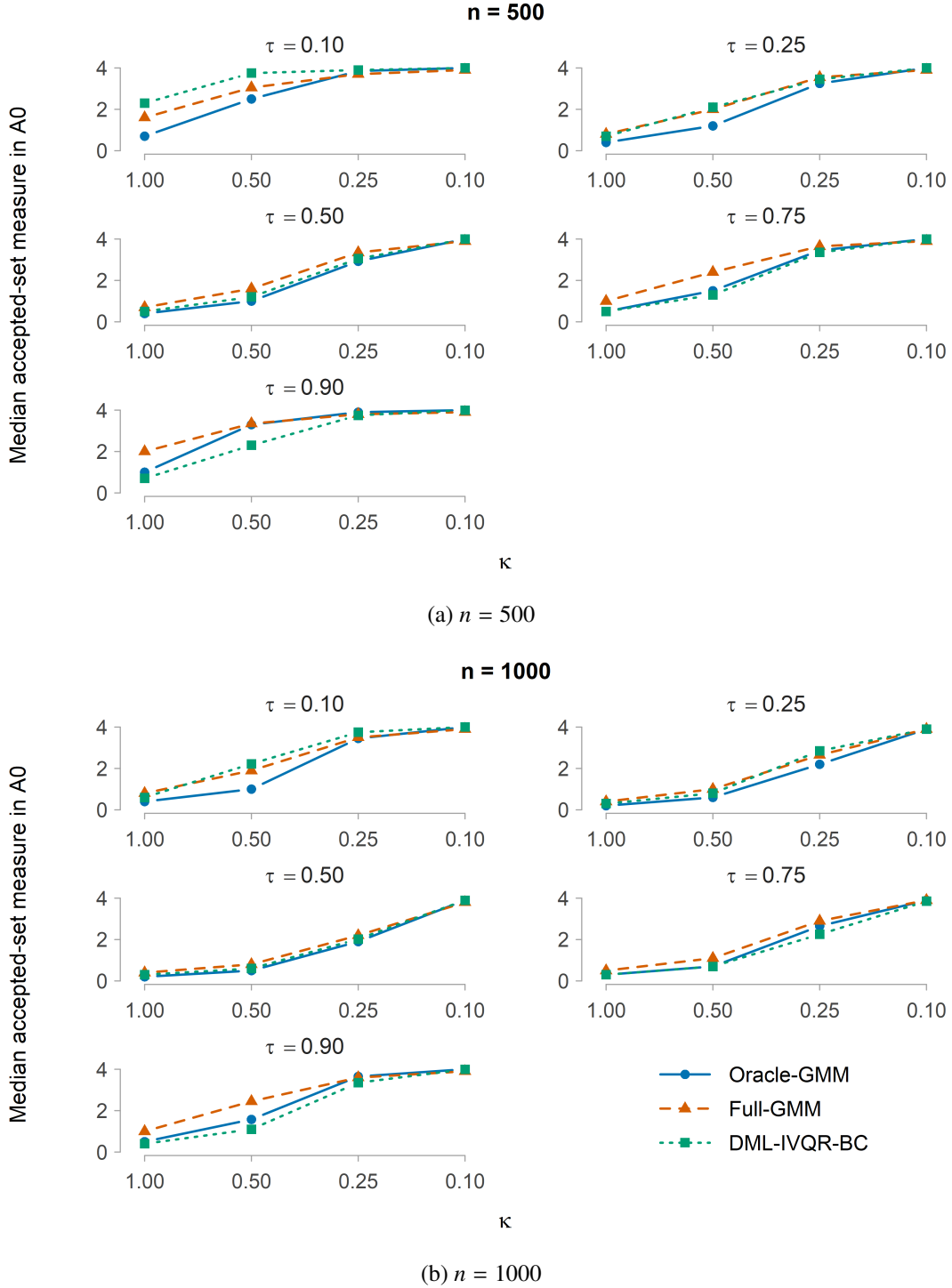


Figure 2. Grid-based accepted-set measure within the primary computational domain $A_0 = [-1, 3]$.

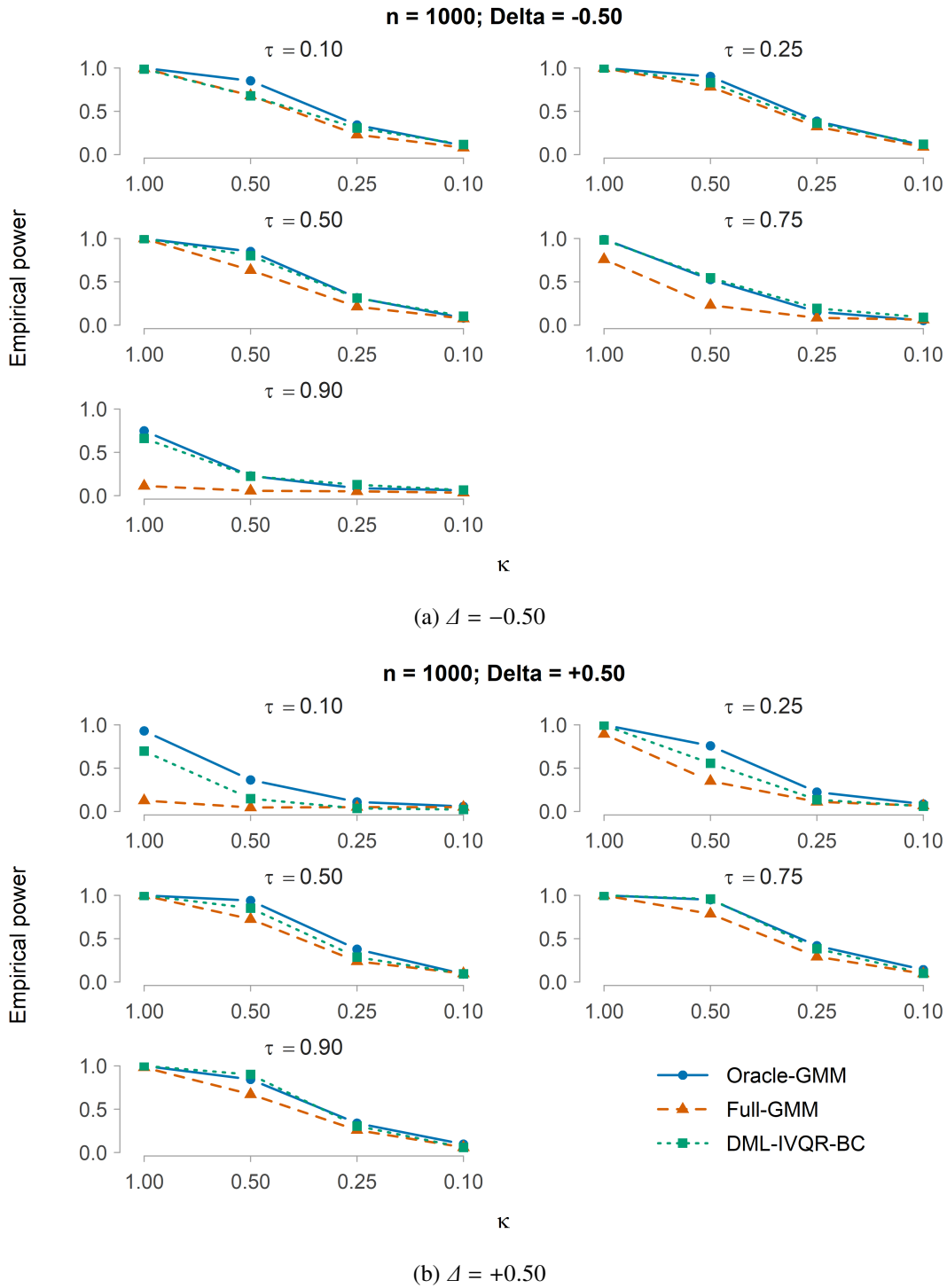


Figure 3. Rejection probability under moderate false alternatives, $n = 1000$.